

Yuxin Xiao

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Research Interest

My current research centers around two primary questions:

- How can we reliably distinguish when language models should incorporate human inputs for effective knowledge updates and when they should resist malicious instructions, improving the alignment and ethics of these tools in real-world applications?
- How can we leverage uncertainty quantification, mechanistic interpretations, and reasoning capabilities of language models to effectively assess, understand, and align their behaviors for trustworthy deployment, especially in high-stakes scenarios?

Education

Massachusetts Institute of Technology (MIT)

PH.D. IN SOCIAL & ENGINEERING SYSTEMS AND STATISTICS

Cambridge, MA

09/2022 - Present

- Advised by **Prof. Marzyeh Ghassemi**; GPA: 5.00/5.00
- Affiliated with Institute for Data, Systems, and Society (IDSS) and Laboratory for Information and Decision Systems (LIDS)

Carnegie Mellon University (CMU)

M.S. IN MACHINE LEARNING

Pittsburgh, PA

08/2020 - 12/2021

- Advised by **Prof. Eric Xing** and **Prof. Louis-Philippe Morency**; GPA: 4.12/4.33

University of Illinois at Urbana-Champaign (UIUC)

B.S. IN COMPUTER SCIENCE; B.S. IN STATISTICS, MATHEMATICS

Urbana, IL

08/2016 - 05/2020

- Advised by **Prof. Jiawei Han** and **Prof. Hari Sundaram**; GPA: 3.93/4.00

Awards & Honors

2020 **C. W. Gear Outstanding Undergraduate Award**, UIUC (2 at UIUC)

2020 **CRA Outstanding Undergraduate Researcher Award (Honorable Mention)**, CRA (4 at UIUC)

Publications

(* indicates equal contribution)

When Style Breaks Safety: Defending Language Models Against Superficial Style Alignment

YUXIN XIAO, SANA TONEKABONI, WALTER GERYCH, VINITH SURIYAKUMAR, MARZYEH GHASSEMI

Preprint 2025

[Paper], [Code]

KScope: A Framework for Characterizing the Knowledge Status of Language Models

YUXIN XIAO, SHAN CHEN, JACK GALLIFANT, DANIELLE BITTERMAN, THOMAS HARTVIGSEN, MARZYEH GHASSEMI

Preprint 2025

[Paper], [Code]

Speak Easy: Eliciting Harmful Jailbreaks from LLMs with Simple Interactions

YIK SIU CHAN*, NARUTATSU RI*, **YUXIN XIAO***, MARZYEH GHASSEMI (poster presentation)

ICML 2025

[Paper], [Code]

SFTMix: Elevating Language Model Instruction Tuning with Mixup Recipe

YUXIN XIAO, SHUJIAN ZHANG, WENXUAN ZHOU, MARZYEH GHASSEMI, SANQIANG ZHAO

Preprint 2024

[Paper], [Code]

In the Name of Fairness: Assessing the Bias in Clinical Record De-identification

YUXIN XIAO*, SHULAMMITE LIM*, TOM JOSEPH POLLARD, MARZYEH GHASSEMI (oral presentation)

FACCT 2023

[Paper], [Code]

Uncertainty Quantification with Pre-trained Language Models: A Large-Scale Empirical Analysis

YUXIN XIAO, PAUL PU LIANG, UMANG BHATT, WILLIE NEISWANGER, RUSLAN SALAKHUTDINOV, LOUIS-PHILIPPE MORENCY (findings)

EMNLP 2022

[Paper], [Code]

SAIS: Supervising and Augmenting Intermediate Steps for Document-Level Relation Extraction

YUXIN XIAO, ZECHENG ZHANG, YUNING MAO, CARL YANG, JIAWEI HAN (oral presentation)

NAACL 2022

[Paper], [Code]

Amortized Auto-Tuning: Cost-Efficient Bayesian Transfer Optimization for Hyperparameter Recommendation

YUXIN XIAO, ERIC P. XING, WILLIE NEISWANGER

Preprint 2021

[Paper], [Code]

Heterogeneous Network Representation Learning: A Unified Framework with Survey and Benchmark

CARL YANG*, **YUXIN XIAO***, YU ZHANG*, YIZHOU SUN, JIAWEI HAN (380+ citations, 300+ GitHub stars and forks)

TKDE 2020

[Paper], [Code]

Discovering Strategic Behaviors for Collaborative Content-Production in Social Networks

YUXIN XIAO, ADIT KRISHNAN, HARI SUNDARAM (oral presentation)

WWW 2020

[Paper], [Code]

Non-local Attention Learning on Large Heterogeneous Information Networks

YUXIN XIAO*, ZECHENG ZHANG*, CARL YANG, CHENGXIANG ZHAI (oral presentation)

IEEE BigData 2019

[Paper], [Code]

Industry Experience

Microsoft Research RESEARCH INTERN, AI ARCHITECTURE & STRATEGY TEAM

Redmond, WA 06/2025 - 08/2025

Zoom RESEARCH INTERN, GENAI R&D TEAM

San Jose, CA 06/2024 - 08/2024

Bosch Center for AI RESEARCH INTERN, NLP TEAM

Renningen, Germany 06/2023 - 08/2023